

MODULE 1 · WEEK 1

Setting Up Your Tech Stack

Git, VS Code, Python & GitHub — the professional environment you'll use for the rest of the course (and your career).

Lectures 69–70 · Welcome + your first repos



Welcome to OPIM 5512

- This course teaches data science **the way it's actually done** — clean Python scripts, version control, the cloud.
- No Google Drive mounts, no Colab-only tricks. Everything runs from your own machine and lives on GitHub.
- Put the time in, show up, and you'll leave able to build and ship real ML projects.

Key idea The goal isn't just to learn models — it's to work like a professional data scientist.



This week

- 1 **Why the tech stack matters** — setup is a real job skill, not busywork
- 2 **The tools** — Git, VS Code, Python, virtual environments
- 3 **GitHub** — account, clone, commit, push
- 4 **Your first repo** — Assignment 01 (A01)
- 5 **Virtual environments** — keep every project isolated



Why setup matters

- In your first job you'll spend real time getting an environment working — and doing it **without asking for help** signals competence.
- Every team has a "stack": the agreed-upon set of tools everyone uses.
- Git, VS Code, and virtual environments are used by data scientists at **every major company**. Learn them once, use them forever.

The core tools

- **Git** — version control: track every change, never lose work, collaborate safely.
- **VS Code** — the editor where you'll write and run your Python.
- **Python** — the language for the whole course.
- **GitHub** — the cloud home for your code.

```
git --version      # confirm Git is installed
python --version   # confirm Python is installed
```

The Git workflow you'll repeat forever

- **Clone** a repo to your machine, **make a change**, then **push** it back to GitHub.
- Three commands cover 90% of daily work: **add** → **commit** → **push**.
- Commit early, commit often — each commit is a save point you can return to.

```
git clone https://github.com/you/opim5512-a01.git
git add .
git commit -m "my first change"
git push
```

Key idea Add, commit, push — this loop is the heartbeat of every real project.

Virtual environments

- A **venv** gives each project its own isolated set of packages so they don't step on each other.
- Create it once per module, activate it whenever you work, install from `requirements.txt`.
- This is what kills "but it works on my machine" — everyone's environment matches.

```
python -m venv .venv
.venv\Scripts\activate      # Windows
pip install -r requirements.txt
```



Your first repo (A01)

- Create your personal class repo: `opim5512-<your-netid>`.
- Make your first repos from scratch and push them — this is how you'll hand in work all semester.
- Getting green check marks on A01 means your whole stack is wired up correctly.



Lab this week

- Work through the setup guide top-to-bottom, then create your class repo and complete A01.
- Goal: run `git --version` and `python --version` with no errors, and push a repo to GitHub.

In the labs repo: `Module1/Week1_TechStack/setup_guide.md` · `Module1/Week1_TechStack/github_fundamentals.md`



Key takeaways

- A working environment is the foundation of all data science — invest in it now.
- Git + GitHub = version control and a cloud home for your code (clone → add → commit → push).
- Virtual environments keep projects isolated and reproducible.
- Finish A01 and you've proven your full stack works end-to-end.